

Supplementary Examination – Summer 2022

Course :B.Tech.

Branch : Civil Engg.

Semester : III

Subject Code & Name : BTCVC304 Hydraulics-1

Max Marks : 60

Date : / /

Duration: 3 Hr.

**Instructions to the Students:**

1. All the questions are compulsory.
2. The level of question/expected answer as per OBE or the Course Outcome (CO) on which the question is based is mentioned in ( ) in front of the question.
3. Use of non-programmable scientific calculators is allowed.
4. Assume suitable data wherever necessary and mention it clearly.

|                                                                                                                                                                                                                                                                                                                     | (Level/CO) | Marks |
|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------|-------|
| <b>Q.1 Solve Any Two of the Following</b>                                                                                                                                                                                                                                                                           |            |       |
| A) Explain the terms specific volume, specific gravity and density.                                                                                                                                                                                                                                                 | CO2        | 06    |
| B) State the newton's law of viscosity and give examples of its applications.                                                                                                                                                                                                                                       | CO2        | 06    |
| C) Calculate the density, specific weight and weight of one liter of petrol of specific gravity =0.7                                                                                                                                                                                                                | CO2        | 06    |
| <b>Q.2 Solve Any Two of the Following</b>                                                                                                                                                                                                                                                                           |            |       |
| A) Explain flow net and its uses?                                                                                                                                                                                                                                                                                   | CO2        | 06    |
| B) Derive Euler's equation of motion?                                                                                                                                                                                                                                                                               | CO2        | 06    |
| C) A 30 cm diameter pipe conveying water, branches into two pipe of diameter 20 cm and 15 cm respectively. If the average velocity in the 30 cm diameter pipe is 2.5 m/s. Find the discharge in this pipe, also determine the velocity in the 15 cm pipe if the average velocity in a 20 cm diameter pipe is 2 m/s? | CO3        | 06    |
| <b>Q.3 Solve Any Two of the Following</b>                                                                                                                                                                                                                                                                           |            |       |
| A) Derive fully developed laminar flow between infinite parallel plate when both plates are stationary?                                                                                                                                                                                                             | CO1        | 06    |
| B) Explain Prandtl mixing length theory?                                                                                                                                                                                                                                                                            | CO2        | 06    |
| C) What is the boundary layer? Explain in detail.                                                                                                                                                                                                                                                                   | CO2        | 06    |
| <b>Q.4 Solve Any two of the Following</b>                                                                                                                                                                                                                                                                           |            |       |
| A) Derive Darcy-Weisbach equation.                                                                                                                                                                                                                                                                                  | CO2        | 06    |
| B) Explain the terms prototype, model analysis and hydraulic similitude.                                                                                                                                                                                                                                            | CO4        | 06    |
| C) State and explain Buckingham's Pi theorem.                                                                                                                                                                                                                                                                       | CO2        | 06    |

**Q.5 Solve Any Two of the Following**

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|----|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|----|
| A) | What is major loss in flow through pipe and what causes the major loss ?<br>Derive the equation for the major loss.                                                                                                              | CO3 | 06 |
| B) | Three pipes of lengths 800m, 500m and 400m and of diameters 500 mm, 400mm and 300mm respectively are connected in series. These pipes are to be replaced by single pipes of length 1700 m. Find the diameter of the single pipe. | CO3 | 06 |
| C) | Define and explain the terms : (i) Hydraulic gradient line and (ii) Total energy line                                                                                                                                            | CO1 | 06 |